1-what is the key limitation of propositional logic for modeling domains like the Wumpus World?

A) lack of consistency
B) no support for negation
C) inability to express general rules compactly
D) absence of backward chaining

2-what does it mean for an inference procedure to be complete?

A) it returns an answer in polynomial time
B) it can prove all tautologies
C) it can derive every logically entailed conclusion from a knowledge base
D) it never produces contradictions

3-what is a model in classical logic?

A) a set of syntactic inference rules making a sentence true
B) a CNF transformation of a sentence that makes it true
C) an interpretation under which a sentence or set of sentences is true
D) a minimal satisfying assignment for a set of literals

4-what is the central strategy in backward chaining?

A) start with all known facts and derive new ones
B) enumerate all possible truth assignments
C) start with what needs to be proven and work towards the premises trying to show if they are true
D) randomly generate models to test entailment

5-which of the following is logically entailed by a knowledge base only if the inference system is complete?

A) a contradiction
B) a tautology
C) a non-trivial consequence
D) the syntax for all rules

6-what kind of clauses are used in forward chaining algorithm?

A) biconditional clauses
B) CNF clauses
C) definite clauses
D) negated implications

7-why do resolution algorithms use conjunctive normal form CNF?

A) it is more human readable
B) it standardizes input for applying the resolution rule
C) it avoids the use of implications
D) it reduces the number of models

8-which of the following is a well-formed term in first order logic?

A) $\exists x\ (P(x) \wedge Q(x))$
B) $f(g(x),y)$
C) $\neg \forall x\ Q(x)$
D) $\forall x \rightarrow y$

9-which of the following violates the syntax rules of FOL?

A) $\forall\ \exists x$: Likes (x, IceCream)
B) $\neg \forall x$ Mortal (x)
C) $\exists x\ \exists y$ Parent (x,y)
D) $\forall x\ \exists y$ Loves (x,y)

10-which task requires knowledge engineering to formalize in FOL?

A) truth table construction
B) CNF conversion
C) encoding real world relationships like kinship
D) elimination tautologies

11-what is a known limitation of unrestricted expressiveness in FOL?

A) it limits recursive definitions
B) it can make inference undecidable or intractable
C) it restricts the use of constants
D) it prevents symbolic reasoning

12-what makes a sentence in FOL “complex”?

A) it contains constants
B) it contains variables
C) it includes logical connectives or quantifiers
D) it uses predicates